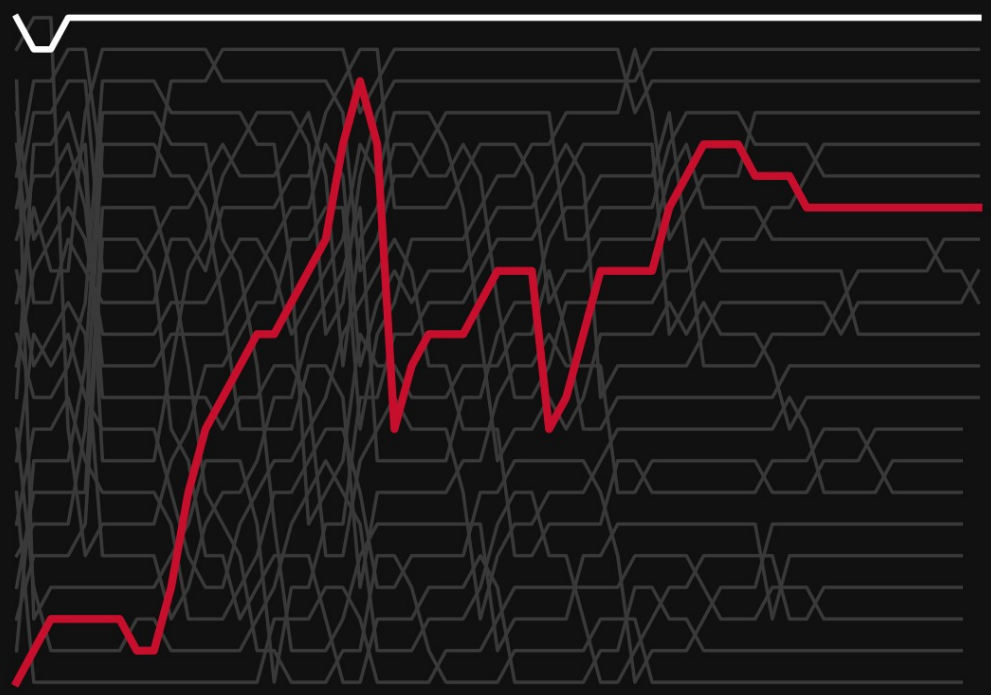
FINAL REVIEW · DATA VISUALISATION CASE STUDY

Formula 1, read as a chain of cause and effect



Drivers, constructors, race strategy and performance trends, 2015–2024

Arwindh C N
CB.SC.U4CSE24207 · 23CSE353 Data Visualisation

209
races analysed

10
seasons, 2015–2024

227,123
lap-by-lap records

15 + 1
visualizations + dashboard

Why do races end the way they do? Six questions, one story

Problem background

A Formula 1 result depends on many linked factors: the car, the driver, qualifying, pit stops, the track and reliability. A results table shows who won, not why.

Problem statement

Which factors really shape a race result, by how much, and which only appear to?

Why it matters

A strategist has seconds to decide. Knowing which levers are real, and which are myths, changes the call.

Approach

One story in five acts, told to one reader: the strategist on the pit wall.

Analytical objectives (from the Primary Review)

O1	Driver performance across seasons	Act 4
O2	Constructor performance across seasons	Act 4
O3	Qualifying position vs final position	Act 1
O4	Pit-stop strategy vs race outcome	Acts 2–3
O5	Driver performance across circuits	Act 4
O6	DNF and retirement patterns over time	Act 4

Ten seasons, 209 races, every car on every lap

Source

Ergast-lineage Formula 1 dataset (the data behind the Jolpica F1 API): 14 linked CSV tables, 1950–2024.

Coverage

2015–2024: 209 races, 53 drivers, 18 constructors, 32 circuits.

Data quality concerns

641 drivers have no finishing position; "+1 Lap" looks like a retirement but is not; 79 pit-lane starts are coded as grid 0; missing values are written as \N.

Table	Rows	One row is
results	4,219	one driver in one race
qualifying	4,205	one driver's qualifying session
pit_stops	7,544	one pit stop (lap, pit-lane time)
lap_times	227,123	one car on one lap (position, time)
standings (final)	223 / 101	a driver's / team's final season table
races, circuits	209 / 32	calendar and track details

All tables link through ID numbers (raceId, driverId, constructorId), so the joins are exact even when a team changes its name.

Fourteen raw tables cleaned into one driver-race dataset

1. 14 raw CSVs

1950–2024; missing values written as \N

2. Keep 2015–2024

209 of 1,125 races

3. Join on IDs

never on names: teams rename (Toro Rosso → RB)

4. Fix and flag

grid 0 → missing; lapped car ≠ DNF

5. Merge and check

one driver-race table; every count = workbook

Before → after	Raw	Clean
Race-result rows	26,759	4,219
Races	1,125	209
Lap records	589,081	227,123
Columns (results → merged)	18	35
Pit-stop time	text (m:ss)	seconds

Key decisions

Two flags, not one: 641 unclassified vs 715 genuine DNFs. 74 cars retired late but still got a position.

Pit time from milliseconds: 484 stops of 60 s or more (red flags, repairs); normal median 23.7 s.

Verified: every row count matches the Primary Review workbook.

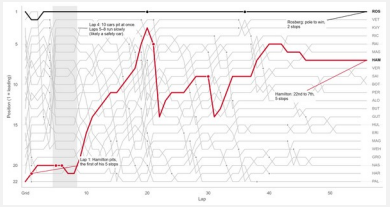
Three tools, each doing the job it does best

Matplotlib

Precise, hand-annotated story charts

V1 bump chart · V4 slope chart · V10 DNF trend · V11 wins bars · V15 misleading pair

Exact control of every mark, label and axis, which the ethics demo needs.

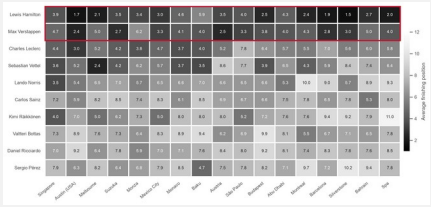


Seaborn

Statistics with confidence ranges

V2 density · V3 lollipop · V5 strip plot · V6–V8 point plots · V13 heatmap · V14 regression

Bootstrap confidence intervals and grouping in one call.

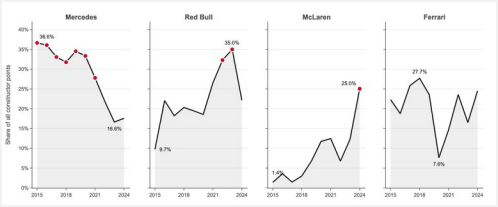


Plotly

Interaction

V9 points-share eras · V12 season points · the five-act dashboard

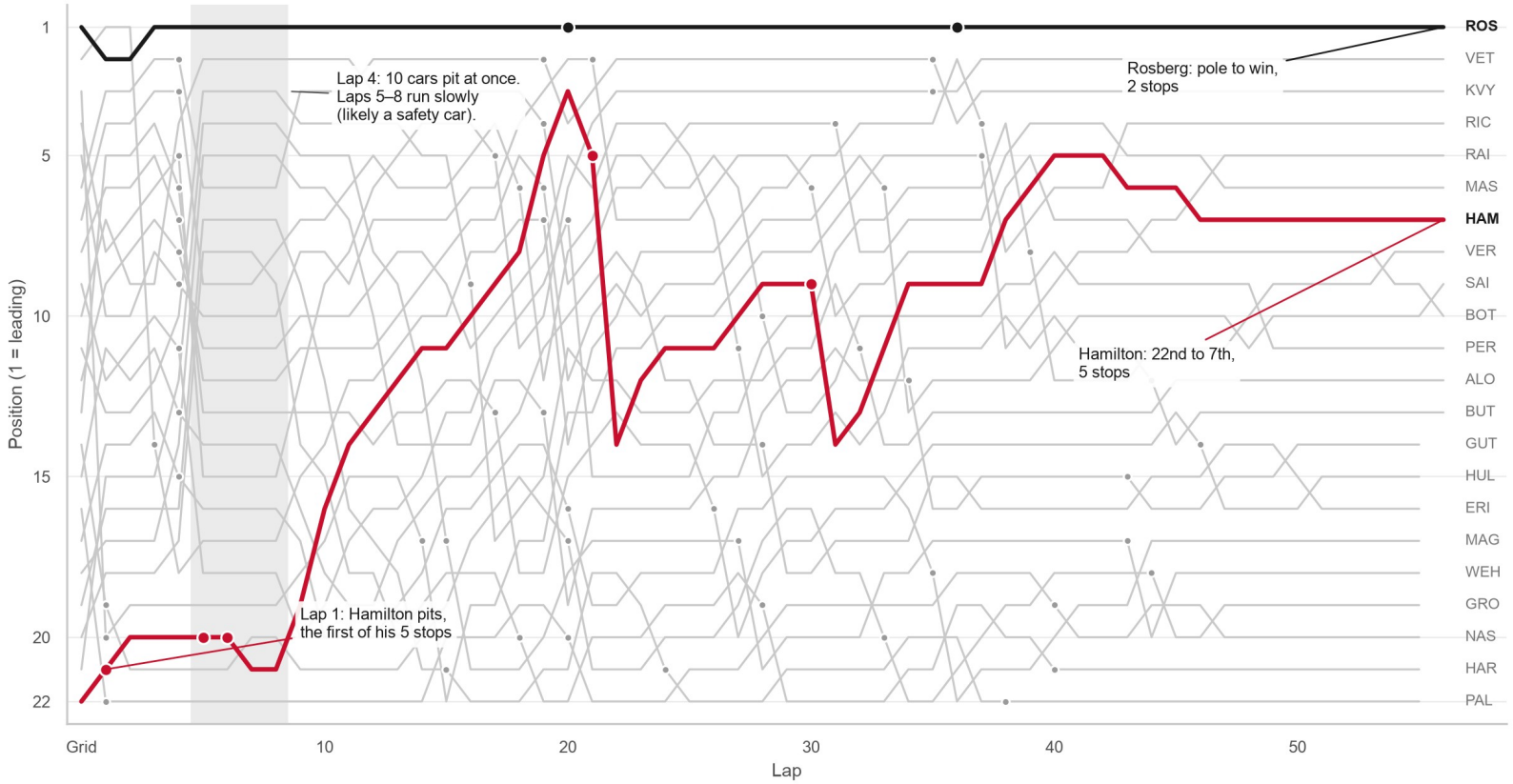
Hover, filters and one offline HTML file anyone can explore.



Beyond basic charts: bump chart, lollipop with 95% ranges, slope chart, dodged point plots, small multiples, annotated heatmap, misleading-vs-honest pair and a linked dashboard. One colour rule throughout: greys for context, red for the thing to look at.

Plotly replaced Power BI so the whole analysis stays reproducible in a single notebook.

844 position changes in one race, and the pole-sitter still won



Visualization 1 · Bump chart with pit-stop markers · Matplotlib

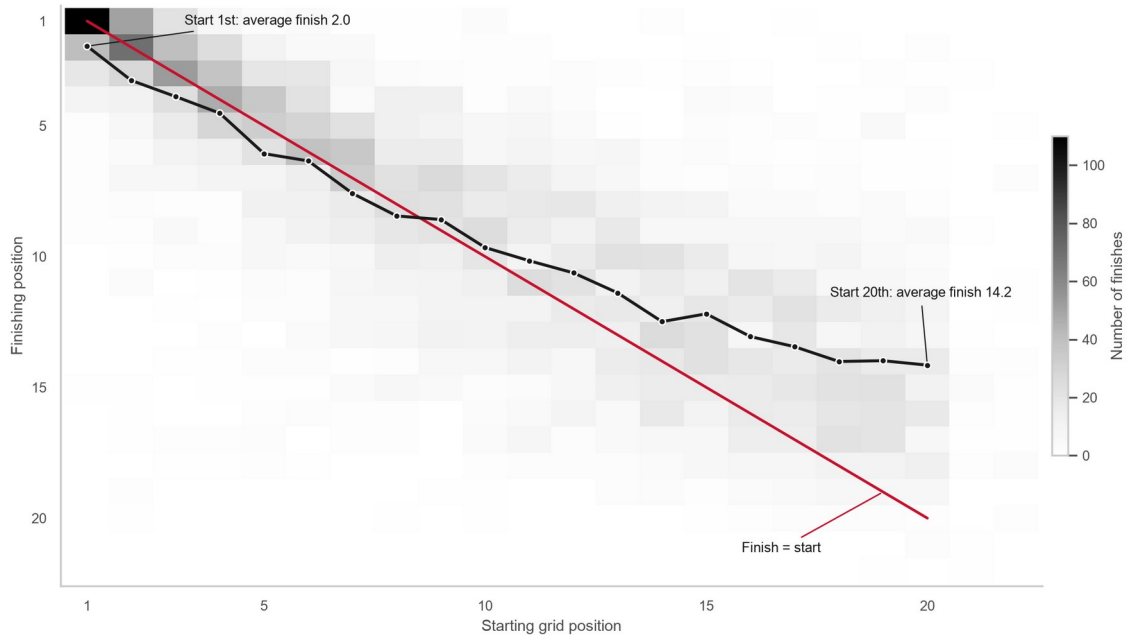
Observation

The 2016 Chinese GP had the most position changes of all 209 races. Hamilton (red) started 22nd, made 5 stops and finished 7th. Rosberg (black), same car, won from pole.

The call you would make

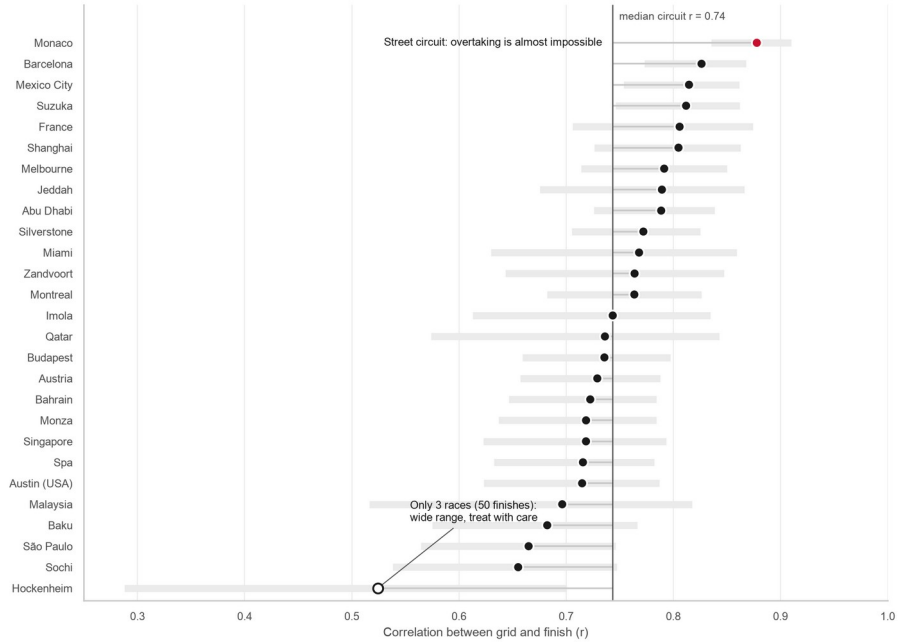
Do not read a race from the result. Read it from the moves.

Saturday decides more than you think: Monaco locks the order in



Visualization 2 · 2D histogram with mean line · Seaborn

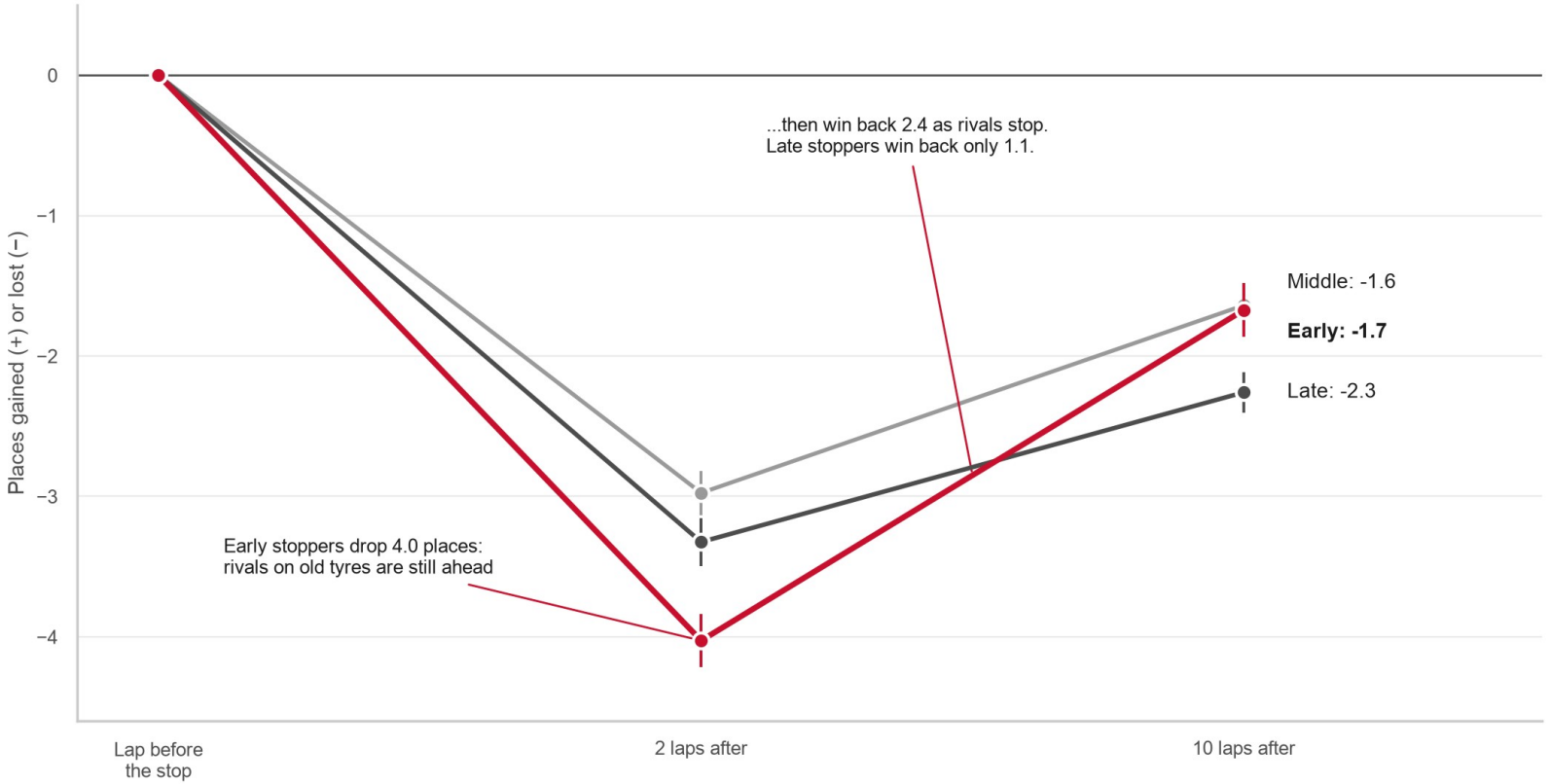
Observation: strong link between grid and finish (r = 0.75); pole won 53% of races.



Visualization 3 · Lollipop with 95% ranges · Seaborn

Observation: Monaco 0.88; Hockenheim 0.52, but only 3 races, so its range is wide.

Early stoppers lose 4 places, then win back 2.4 of them



Visualization 4 · Slope chart with error bars · Matplotlib

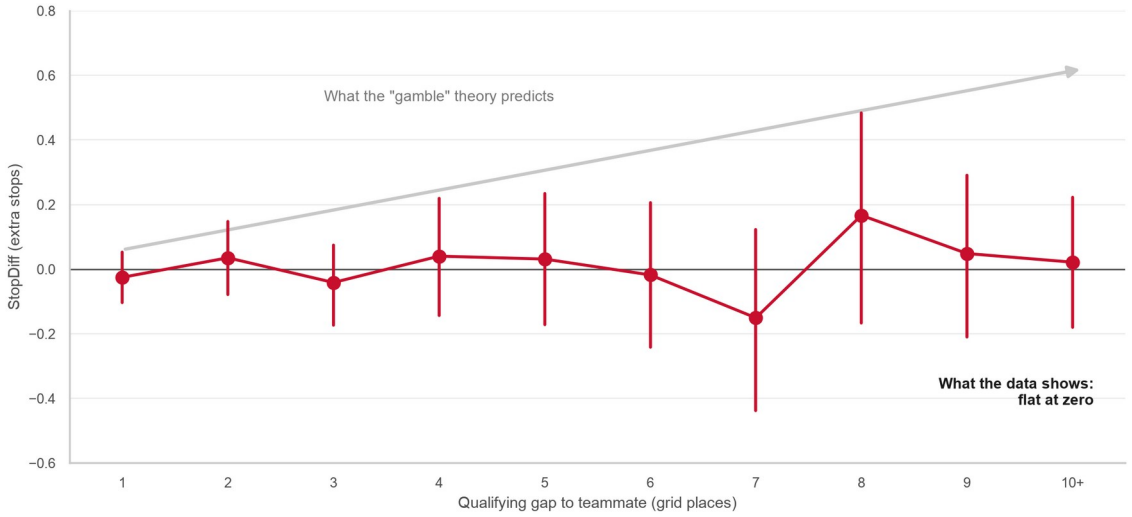
Observation

Across 3,490 first pit stops, early stoppers go -4.0 then -1.7; late stoppers -3.3 then -2.3. Same pattern without the stops in laps 1–5.

The call you would make

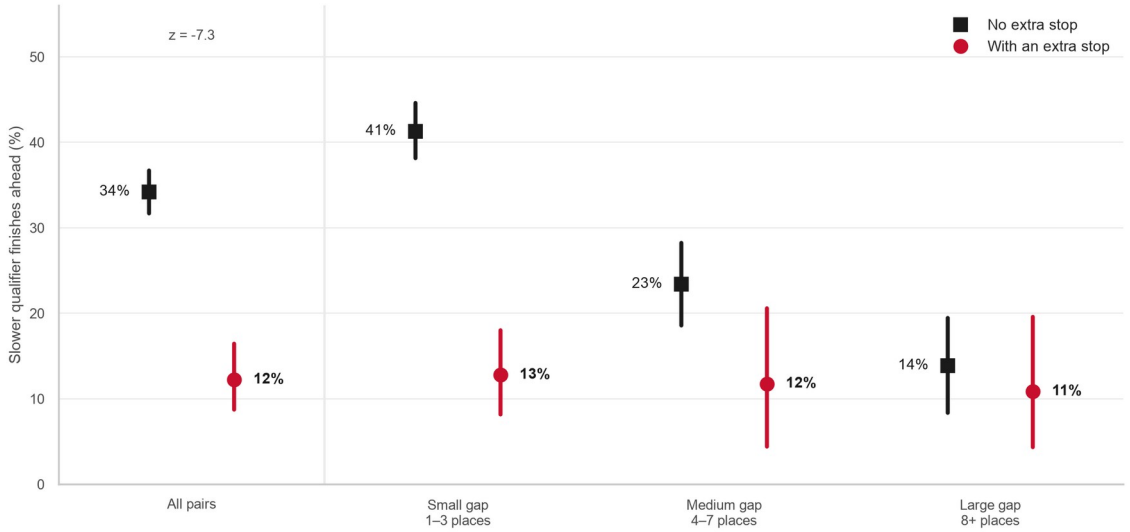
If the car can survive two or three exposed laps, stop first.

The myth is dead, the alarm is real: an extra stop means trouble



Visualization 6 · Point plot vs expected trend · Seaborn

Observation: the myth says drivers far behind their teammate gamble on extra stops. No link: r = +0.02.

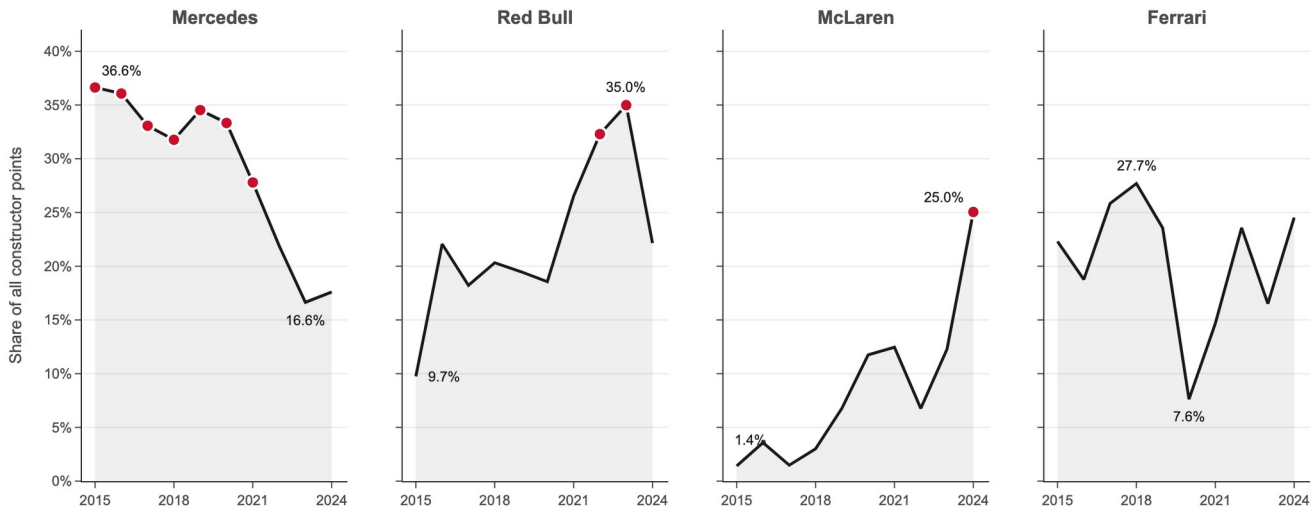


Visualization 7 · Dodged point plot with CIs · Seaborn

Observation: with an extra stop the slower teammate wins 12% of duels, not 34% (z = -7.3).

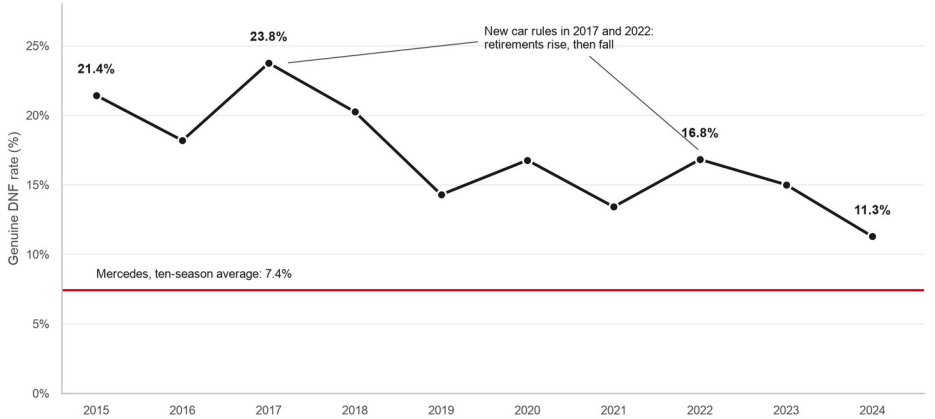
The call you would make: treat an unplanned stop as an alarm, not a strategy, and protect the other car's result.

The car sets the ceiling: three eras in ten seasons



Visualization 9 · Small-multiple area charts · Plotly

Observation: Mercedes 36.6% of points (2015) → 16.6% (2023); Red Bull up to 35.0%; McLaren 1.4% → 25.0%.

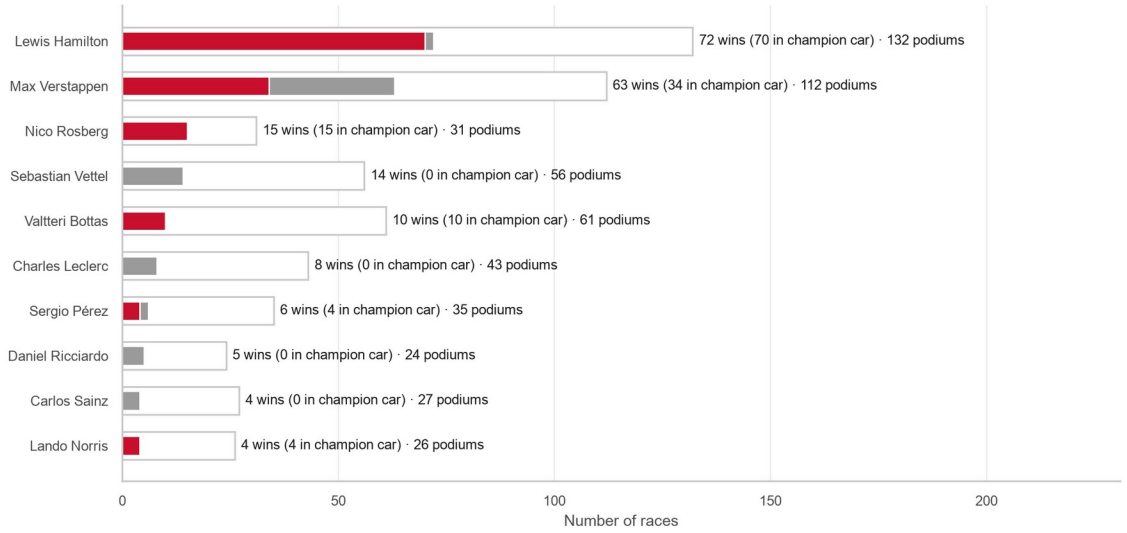


Visualization 10 · Trend line with reference line · Matplotlib

Observation: genuine DNFs 21.4% → 11.3%; Mercedes 7.4%.

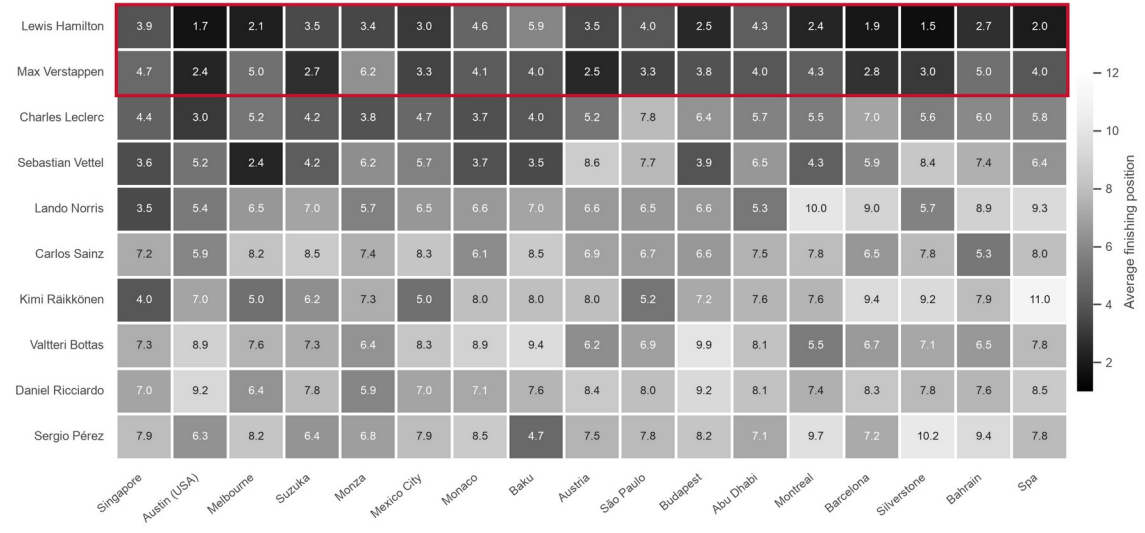
The call you would make: plan the season around the car: every edge from Acts 1–3 lives inside its ceiling.

Wins follow the car, and the driver matters more than the track



Visualization 11 · Stacked bars with podium outline · Matplotlib

Observation: 67% of all wins went to that season's champion team; Hamilton 70 of 72 in the champion car.

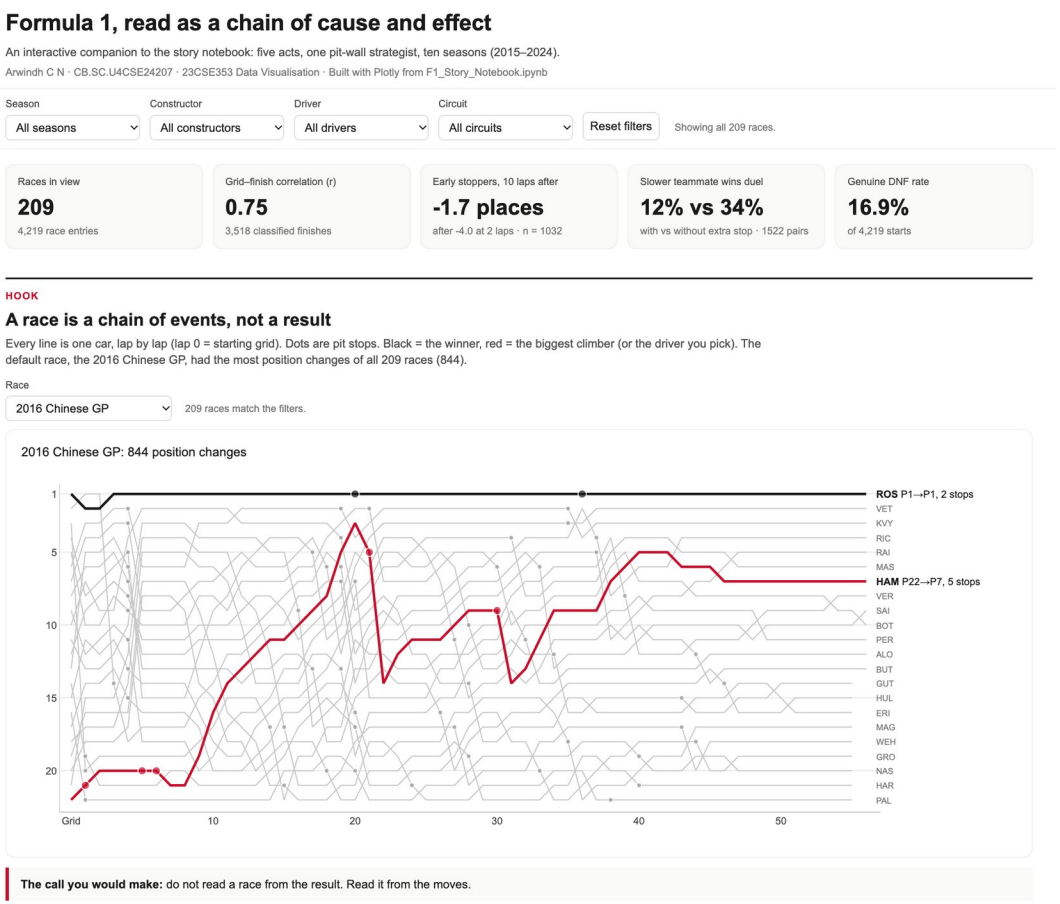


Visualization 13 · Annotated heatmap · Seaborn

Observation: driver rows differ by 4.6 places, circuit columns by only 1.7.

Honest anti-climax: lap-time consistency barely predicts points (r = -0.18, Visualization 14), and other methods give +0.03 to -0.40.

One interactive page: four filters, five acts, every chart linked



Components: 5 headline tiles (races, grid–finish r, early-stopper change, teammate duel, DNF rate), a race selector with the lap-by-lap chart, and 8 linked charts arranged as the five acts.

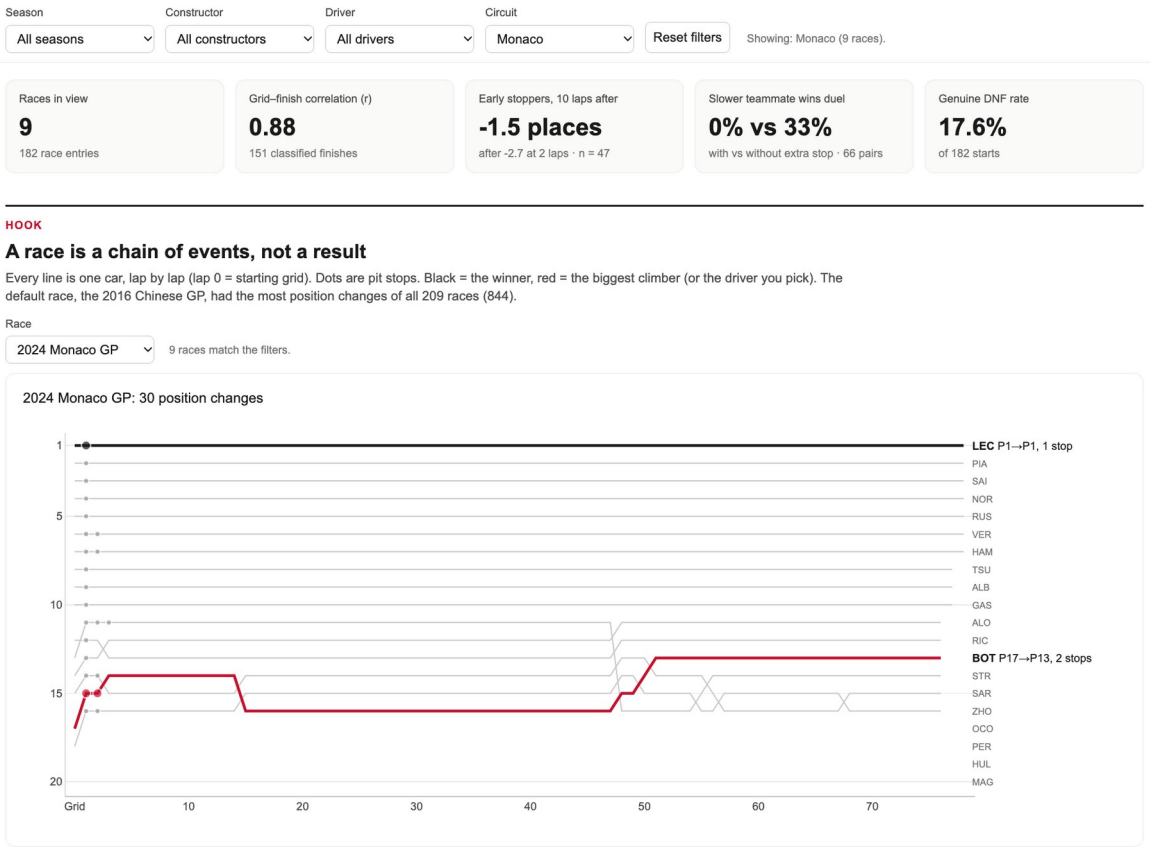
Filters: season, constructor, driver and circuit in one row; every chart re-draws for the same selection.

Interactions: hover for exact values; the chosen team or driver turns red; a toggle switches the Act 5 chart between misleading and honest; the view is saved in the link.

Honesty built in: small selections are labelled "small sample"; grey lines show the all-data result behind a filtered one.

Delivery: one HTML file (5.9 MB) that works offline, generated by the notebook.

Filter to Monaco and the story sharpens: r rises from 0.75 to 0.88



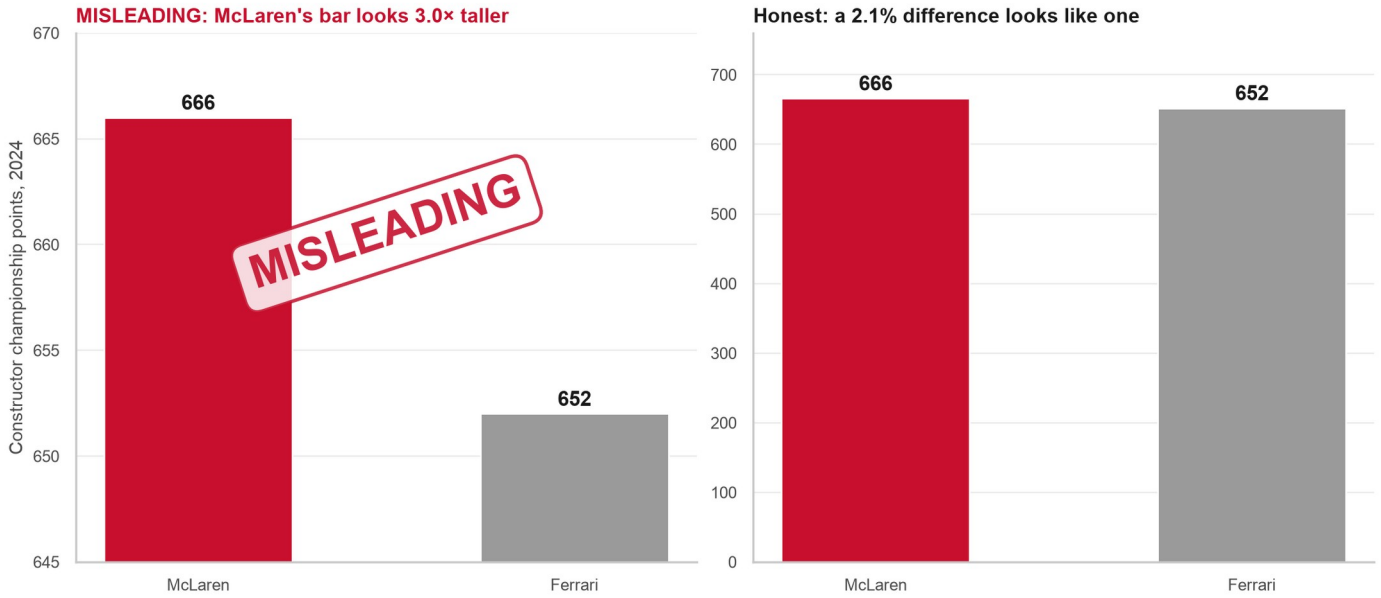
KPI	All races	Circuit = Monaco
Races	209	9
Grid–finish correlation r	0.75	0.88
Early stoppers after 10 laps	-1.7	-1.5 (n = 47)
Slower teammate wins (extra vs none)	12% vs 34%	0% vs 33% (only 11 extra)
Genuine DNF rate	16.9%	17.6%

What? At Monaco the start order mostly holds; the 2024 Monaco GP had 30 position changes against 844 in the hook race.

So what? At street circuits the race is largely decided on Saturday.

Now what? All-in on qualifying at Monaco; keep risk for overtaking tracks.

Same numbers, two stories: a cut axis turns a 2.1% gap into a bar 3.0× taller



Visualization 15 · Misleading vs corrected bar pair · Matplotlib

Misleading chart

Y-axis starts at 645, not 0. The real numbers (McLaren 666, Ferrari 652) look like a rout.

False impression

"McLaren crushed Ferrari." In fact it was the closest constructors' title of the decade: 14 points.

Corrected chart and principle

Zero baseline and values on the bars. Truthful scaling and proportional ink: lie factor about 93 → 1.

Two expectations were overturned: extra stops and circuit specialists

Objective	What the data shows	As expected?	Visual
O1 Drivers across seasons	Hamilton 72 wins, Verstappen 63; titles follow the car eras	Partly	V11, V12
O2 Constructors across seasons	Three eras: Mercedes 36.6% (2015) → Red Bull 35.0% (2023) → McLaren champion 2024	Yes	V9
O3 Qualifying vs finish	Strong link, r = 0.75; Monaco 0.88, Hockenheim 0.52	Yes, varies by track	V2, V3
O4 Pit strategy vs outcome	Early stops win back 2.4 places vs 1.1; extra stop: 12% vs 34%	No: a symptom, not a gamble	V4, V6, V7
O5 Drivers across circuits	Driver rows differ by 4.6 places, circuits by 1.7	No: specialists are rare	V13
O6 DNFs over time	Genuine DNFs fell from 21.4% to 11.3%	Yes	V10

Every result is a correlation in historical records, not an experiment.

The 844 position changes were a chain of cause and effect

Recommendations for the pit wall

1. Weight qualifying effort by circuit: all-in at Monaco, Barcelona and Mexico City.

2. If the car can survive a few exposed laps, stop first: early stoppers win back 2.4 places, late ones 1.1.

3. Treat an unplanned extra stop as an alarm and protect the other car's result.

4. Plan the season around the car: 67% of wins went to the champion team.

Main conclusion

Saturday sets the order, pit timing trades places, an extra stop signals trouble, and the car sets the ceiling.

Limitations

Correlation, not causation. No tyre, weather or safety-car data. Small samples are flagged (Hockenheim: 3 races).

Back to the hook: Rosberg won from pole, 10 cars pitted on lap 4, Hamilton's 3 extra stops were the alarm, and the 2016 Mercedes took 36.1% of all points.